Lab Sheet 6: MongoDB Basic commands

1.
- Create a database called ‘vehicles’ and write a MongoDB query to select database as

“vehicles”.

- Write a MongoDB query to display all the databases.
- Create a collection called ‘two_wheelers’. (use capping) and Create a collection called

‘four_wheelers’.

```
test> show dbs
Animals     80.00 KiB
admin       40.00 KiB
config     108.00 KiB
local       40.00 KiB
vehicles   112.00 KiB
```

```
test> use vehicles
switched to db vehicles
vehicles> show collections
four_wheelers
two_wheelers
```

- Write a MongoDB query to display all documents available in two_wheelers and

four_wheelers

```
vehicles> db.two_wheelers.find()
[
  {
    _id: ObjectId('69b149918af99a0b3f4e2bf6'),
    bike_name: 'Activa 6G',
    model: 'gearless',
    category: '125cc',
    colors_available: [ 'red', 'black', 'blue' ],
    manufacturer: 'Honda',
    performance: 8,
    timestamp: ISODate('2020-01-15T00:00:00.000Z'),
    price: 85000
  },
  {
    _id: ObjectId('69b149918af99a0b3f4e2bf7'),
    bike_name: 'Pulsar 150',
    model: 'gear',
    category: '150cc',
    colors_available: [ 'black', 'blue', 'sport red' ],
    manufacturer: 'Bajaj',
    performance: 9,
    timestamp: ISODate('2019-05-10T00:00:00.000Z'),
    price: 110000
  },
  {
    _id: ObjectId('69b149918af99a0b3f4e2bf8'),
    bike_name: 'Splendor Plus',
    model: 'gear',
    category: '100cc',
    colors_available: [ 'black', 'red' ],
    manufacturer: 'Hero',
    performance: 7,
    timestamp: ISODate('2018-07-20T00:00:00.000Z'),
    price: 75000
  },
  {
    _id: ObjectId('69b149918af99a0b3f4e2bf9'),
    bike_name: 'Apache RTR 160',
    model: 'gear',
    category: '150cc',
    colors_available: [ 'red', 'black' ],
    manufacturer: 'TVS',
    performance: 9,
    timestamp: ISODate('2021-03-12T00:00:00.000Z'),
    price: 120000
  },
  {
    _id: ObjectId('69b149918af99a0b3f4e2bfa'),
    bike_name: 'Access 125',
    model: 'gearless',
    category: '125cc',
    colors_available: [ 'blue', 'white', 'black' ],
    manufacturer: 'Suzuki',
    performance: 8,
    timestamp: ISODate('2020-09-18T00:00:00.000Z'),
    price: 90000
  }
]
```

```
vehicles> db.four_wheelers.find()
[
  {
    _id: ObjectId('69b149ef15a96c5bb4187921'),
    vehicle_name: 'Swift',
    model: 'own',
    category: 'car',
    variants: [ 'vxi', 'zxi', 'petrol' ],
    manufacturer: 'Maruti Suzuki',
    performance: 8,
    timestamp: ISODate('2021-02-10T00:00:00.000Z'),
    price: 700000
  },
  {
    _id: ObjectId('69b149ef15a96c5bb4187922'),
    vehicle_name: 'Innova',
    model: 'commercial',
    category: 'car',
    variants: [ 'diesel', 'zxi' ],
    manufacturer: 'Toyota',
    performance: 9,
    timestamp: ISODate('2020-06-15T00:00:00.000Z'),
    price: 1800000
  },
  {
    _id: ObjectId('69b149ef15a96c5bb4187923'),
    vehicle_name: 'Ashok Leyland Truck',
    model: 'commercial',
    category: 'heavy truck',
    variants: [ 'diesel' ],
    manufacturer: 'Ashok Leyland',
    performance: 7,
    timestamp: ISODate('2019-11-12T00:00:00.000Z'),
    price: 2500000
  },
  {
    _id: ObjectId('69b149ef15a96c5bb4187924'),
    vehicle_name: 'Tata Bus',
    model: 'commercial',
    category: 'bus',
    variants: [ 'diesel' ],
    manufacturer: 'Tata',
    performance: 8,
    timestamp: ISODate('2018-04-22T00:00:00.000Z'),
    price: 3000000
  },
  {
    _id: ObjectId('69b149ef15a96c5bb4187925'),
    vehicle_name: 'Mahindra Bolero',
    model: 'own',
    category: 'car',
    variants: [ 'diesel', 'zxi' ],
    manufacturer: 'Mahindra',
    performance: 7,
    timestamp: ISODate('2022-01-05T00:00:00.000Z'),
    price: 950000
  }
]
```

- Write a MongoDB query to display only vehicle name and price in all the collection of

the database

```
vehicles> db.two_wheelers.find({}, {bike_name:1, price:1, _id:0})
[
  { bike_name: 'Activa 6G', price: 85000 },
  { bike_name: 'Pulsar 150', price: 110000 },
  { bike_name: 'Splendor Plus', price: 75000 },
  { bike_name: 'Apache RTR 160', price: 120000 },
  { bike_name: 'Access 125', price: 90000 }
]
```

```
vehicles> db.four_wheelers.find({}, {vehicle_name:1, price:1, _id:0})
[
  { vehicle_name: 'Swift', price: 700000 },
  { vehicle_name: 'Innova', price: 1800000 },
  { vehicle_name: 'Ashok Leyland Truck', price: 2500000 },
  { vehicle_name: 'Tata Bus', price: 3000000 },
  { vehicle_name: 'Mahindra Bolero', price: 950000 }
]
```

- Write a MongoDB query to display two_wheelers from a particular company

```
vehicles> db.two_wheelers.find({manufacturer:"Honda"})
[
  {
    _id: ObjectId('69b149918af99a0b3f4e2bf6'),
    bike_name: 'Activa 6G',
    model: 'gearless',
    category: '125cc',
    colors_available: [ 'red', 'black', 'blue' ],
    manufacturer: 'Honda',
    performance: 8,
    timestamp: ISODate('2020-01-15T00:00:00.000Z'),
    price: 85000
  }
]
```

- Write a MongoDB query to display four_wheelers available in diesel variants

```
vehicles> db.four_wheelers.find({variants:"diesel"})
[
  {
    _id: ObjectId('69b149ef15a96c5bb4187922'),
    vehicle_name: 'Innova',
    model: 'commercial',
    category: 'car',
    variants: [ 'diesel', 'zxi' ],
    manufacturer: 'Toyota',
    performance: 9,
    timestamp: ISODate('2020-06-15T00:00:00.000Z'),
    price: 1800000
  },
  {
    _id: ObjectId('69b149ef15a96c5bb4187923'),
    vehicle_name: 'Ashok Leyland Truck',
    model: 'commercial',
    category: 'heavy truck',
    variants: [ 'diesel' ],
    manufacturer: 'Ashok Leyland',
    performance: 7,
    timestamp: ISODate('2019-11-12T00:00:00.000Z'),
    price: 2500000
  },
  {
    _id: ObjectId('69b149ef15a96c5bb4187924'),
    vehicle_name: 'Tata Bus',
    model: 'commercial',
    category: 'bus',
    variants: [ 'diesel' ],
    manufacturer: 'Tata',
    performance: 8,
    timestamp: ISODate('2018-04-22T00:00:00.000Z'),
    price: 3000000
  },
  {
    _id: ObjectId('69b149ef15a96c5bb4187925'),
    vehicle_name: 'Mahindra Bolero',
    model: 'own',
    category: 'car',
    variants: [ 'diesel', 'zxi' ],
    manufacturer: 'Mahindra',
    performance: 7,
    timestamp: ISODate('2022-01-05T00:00:00.000Z'),
    price: 950000
  }
]
```

- Write a MongoDB query to display vehicles name, category and manufacturer details

whose rating is more than 5.

```
vehicles> db.two_wheelers.find(
| {performance: {$gt:5}},
| {bike_name:1, category:1, manufacturer:1, _id:0}
| )
[
  { bike_name: 'Activa 6G', category: '125cc', manufacturer: 'Honda' },
  { bike_name: 'Pulsar 150', category: '150cc', manufacturer: 'Bajaj' },
  { bike_name: 'Splendor Plus', category: '100cc', manufacturer: 'Hero' },
  { bike_name: 'Apache RTR 160', category: '150cc', manufacturer: 'TVS' },
  { bike_name: 'Access 125', category: '125cc', manufacturer: 'Suzuki' }
]
```

```
vehicles> db.four_wheelers.find(
| {performance: {$gt:5}},
| {vehicle_name:1, category:1, manufacturer:1, _id:0}
| )
[
  {
    vehicle_name: 'Swift',
    category: 'car',
    manufacturer: 'Maruti Suzuki'
  },
  { vehicle_name: 'Innova', category: 'car', manufacturer: 'Toyota' },
  {
    vehicle_name: 'Ashok Leyland Truck',
    category: 'heavy truck',
    manufacturer: 'Ashok Leyland'
  },
  { vehicle_name: 'Tata Bus', category: 'bus', manufacturer: 'Tata' },
  {
    vehicle_name: 'Mahindra Bolero',
    category: 'car',
    manufacturer: 'Mahindra'
  }
]
```

2.

- Create a database called ‘animal’ and write a MongoDB query to select database as

‘animal’.

- Write a MongoDB query to display all the databases.
- Create a collection called ‘wild_animals’.(use capping) and Create a collection called

‘domestic_animals’.

```
test> show dbs
Animals     80.00 KiB
admin       40.00 KiB
config     108.00 KiB
local       40.00 KiB
vehicles   112.00 KiB
```

```
vehicles> use Animals
switched to db Animals
Animals> show collections
Domestic_animals
Wild_animals
Animals>
```

- Write a MongoDB query to display all documents available in wild_animals and domastic_animals.

```
Animals> db.Wild_animals.find()
[
  {
    _id: ObjectId('69b150595b88e57fc9809a49'),
    animal_name: 'Lion',
    nature: 'harm',
    favorite_foods: [ 'meat', 'deer' ],
    care_taker_name: 'Ravi',
    life_span: 14,
    timestamp: ISODate('2022-05-10T00:00:00.000Z'),
    expenses: 50000
  },
  {
    _id: ObjectId('69b150595b88e57fc9809a4a'),
    animal_name: 'Tiger',
    nature: 'harm',
    favorite_foods: [ 'meat', 'rabbit' ],
    care_taker_name: 'Suresh',
    life_span: 15,
    timestamp: ISODate('2021-03-12T00:00:00.000Z'),
    expenses: 55000
  },
  {
    _id: ObjectId('69b150595b88e57fc9809a4b'),
    animal_name: 'Elephant',
    nature: 'harmless',
    favorite_foods: [ 'grass', 'fruits' ],
    care_taker_name: 'Ramesh',
    life_span: 60,
    timestamp: ISODate('2020-06-18T00:00:00.000Z'),
    expenses: 70000
  },
  {
    _id: ObjectId('69b150595b88e57fc9809a4c'),
    animal_name: 'Deer',
    nature: 'harmless',
    favorite_foods: [ 'grass', 'leaves' ],
    care_taker_name: 'Ravi',
    life_span: 20,
    timestamp: ISODate('2023-01-25T00:00:00.000Z'),
    expenses: 20000
  },
  {
    _id: ObjectId('69b150595b88e57fc9809a4d'),
    animal_name: 'Leopard',
    nature: 'harm',
    favorite_foods: [ 'meat', 'rabbit' ],
    care_taker_name: 'Suresh',
    life_span: 12,
    timestamp: ISODate('2022-11-11T00:00:00.000Z'),
    expenses: 45000
  }
]
```

```
Animals> db.Domestic_animals.find()
[
  {
    _id: ObjectId('69b150b253f55da4c9c7d7bd'),
    animal_name: 'Dog',
    gender: 'male',
    favorite_foods: [ 'meat', 'biscuits' ],
    animal_petname: 'Tommy',
    life_span: 12,
    timestamp: ISODate('2021-04-05T00:00:00.000Z'),
    expenses: 15000
  },
  {
    _id: ObjectId('69b150b253f55da4c9c7d7be'),
    animal_name: 'Cat',
    gender: 'female',
    favorite_foods: [ 'milk', 'fish' ],
    animal_petname: 'Kitty',
    life_span: 15,
    timestamp: ISODate('2022-02-11T00:00:00.000Z'),
    expenses: 12000
  },
  {
    _id: ObjectId('69b150b253f55da4c9c7d7bf'),
    animal_name: 'Cow',
    gender: 'female',
    favorite_foods: [ 'grass', 'fodder' ],
    animal_petname: 'Lakshmi',
    life_span: 20,
    timestamp: ISODate('2020-08-20T00:00:00.000Z'),
    expenses: 18000
  },
  {
    _id: ObjectId('69b150b253f55da4c9c7d7c0'),
    animal_name: 'Horse',
    gender: 'male',
    favorite_foods: [ 'grass', 'grains' ],
    animal_petname: 'Raja',
    life_span: 25,
    timestamp: ISODate('2019-09-15T00:00:00.000Z'),
    expenses: 30000
  },
  {
    _id: ObjectId('69b150b253f55da4c9c7d7c1'),
    animal_name: 'Goat',
    gender: 'female',
    favorite_foods: [ 'leaves', 'grass' ],
    animal_petname: 'Chinni',
```

```
    life_span: 10,
    timestamp: ISODate('2023-03-10T00:00:00.000Z'),
```

- Write a MongoDB query to display only animal name and expenses in all the collection of the database

```
Animals> db.Wild_animals.find({}, {animal_name:1, expenses:1, _id:0})
[
  { animal_name: 'Lion', expenses: 50000 },
  { animal_name: 'Tiger', expenses: 55000 },
  { animal_name: 'Elephant', expenses: 70000 },
  { animal_name: 'Deer', expenses: 20000 },
  { animal_name: 'Leopard', expenses: 45000 }
]
```

```
Animals> db.Domestic_animals.find({}, {animal_name:1, expenses:1, _id:0})
[
  { animal_name: 'Dog', expenses: 15000 },
  { animal_name: 'Cat', expenses: 12000 },
  { animal_name: 'Cow', expenses: 18000 },
  { animal_name: 'Horse', expenses: 30000 },
  { animal_name: 'Goat', expenses: 10000 }
]
Animals> |
```

- Write a MongoDB query to display domestic_animals whose life is a particular year

```
Animals> db.Domestic_animals.find({life_span:12})
[
  {
    _id: ObjectId('69b150b253f55da4c9c7d7bd'),
    animal_name: 'Dog',
    gender: 'male',
    favorite_foods: [ 'meat', 'biscuits' ],
    animal_petname: 'Tommy',
    life_span: 12,
    timestamp: ISODate('2021-04-05T00:00:00.000Z'),
    expenses: 15000
  }
]
Animals> |
```

- Write a MongoDB query to display wild_animals available under a particular care_taker

```
Animals> db.Wild_animals.find({care_taker_name:"Ravi"})
[
  {
    _id: ObjectId('69b150595b88e57fc9809a49'),
    animal_name: 'Lion',
    nature: 'harm',
    favorite_foods: [ 'meat', 'deer' ],
    care_taker_name: 'Ravi',
    life_span: 14,
    timestamp: ISODate('2022-05-10T00:00:00.000Z'),
    expenses: 50000
  },
  {
    _id: ObjectId('69b150595b88e57fc9809a4c'),
    animal_name: 'Deer',
    nature: 'harmless',
    favorite_foods: [ 'grass', 'leaves' ],
    care_taker_name: 'Ravi',
    life_span: 20,
    timestamp: ISODate('2023-01-25T00:00:00.000Z'),
    expenses: 20000
  }
]
Animals> |
```

- Write a MongoDB query to display animal name, favorite_foods and expenses details whose lifespan is more than 5 years.

```
Animals> db.Wild_animals.find(
| {life_span:{$gt:5}},
| {animal_name:1,favorite_foods:1,expenses:1,_id:0}
| )
[
  {
    animal_name: 'Lion',
    favorite_foods: [ 'meat', 'deer' ],
    expenses: 50000
  },
  {
    animal_name: 'Tiger',
    favorite_foods: [ 'meat', 'rabbit' ],
    expenses: 55000
  },
  {
    animal_name: 'Elephant',
    favorite_foods: [ 'grass', 'fruits' ],
    expenses: 70000
  },
  {
    animal_name: 'Deer',
    favorite_foods: [ 'grass', 'leaves' ],
    expenses: 20000
  },
  {
    animal_name: 'Leopard',
    favorite_foods: [ 'meat', 'rabbit' ],
    expenses: 45000
  }
]
Animals> |
```

```
Animals> db.Domestic_animals.find(
| {life_span:{$gt:5}},
| {animal_name:1,favorite_foods:1,expenses:1,_id:0}
| )
[
  {
    animal_name: 'Dog',
    favorite_foods: [ 'meat', 'biscuits' ],
    expenses: 15000
  },
  {
    animal_name: 'Cat',
    favorite_foods: [ 'milk', 'fish' ],
    expenses: 12000
  },
  {
    animal_name: 'Cow',
    favorite_foods: [ 'grass', 'fodder' ],
    expenses: 18000
  },
  {
    animal_name: 'Horse',
    favorite_foods: [ 'grass', 'grains' ],
    expenses: 30000
  },
  {
    animal_name: 'Goat',
    favorite_foods: [ 'leaves', 'grass' ],
    expenses: 10000
  }
]
Animals>
```